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CHAPTER 1

INTRODUCTION

1.1 Project Background and Motivation

Local Government Units (LGUs) in the Philippines are mandated under the Republic Act No. 7160, otherwise famously known as Local Government Code of 1991, they exercise the power to generate their own sources of revenue through the levy of people who pay real property taxes, local taxes, business permits and licenses, regulatory fees, and other charges. Paying taxes correctly and on time helps the Local Government Unit of Caloocan secure the funds needed to support public services and development programs. Every peso that goes uncollected, delayed, or lost in a disorganized process is a peso that could have gone to road repair, health services, or disaster response programs that the city needs. The City Treasurer's Office of Caloocan is the main body responsible for these collection activities, and like most treasury offices in the country, it continues to face the usual difficulties in processing many daily transactions.

Through initial observation at the Treasurer's Office, it was noted that although the city already introduced an electronic official receipt system a few years back that automatically issues receipts for tax payment made online, several parts of the revenue collection process are still handled manually. As a result, treasury personnel must check records from different sources to verify a taxpayer account.

A taxpayer who owns both a residential property and a small business, for example, may have to go through two separate verification processes even though both transactions are supposed to fall under the same treasury system. This setup slows down

transaction processing time, especially during peak periods like the start of the year, when property owners rush to pay before the deadline to avail of discounts, or during business permit renewal season, when the office extends its hours to weekends just to accommodate the volume of people lining up.

Every one of these transactions follows a fairly involved process, the taxpayer submits an application with supporting documents at a designated window, the department verifies the details and sometimes has to send someone out for a field inspection to confirm the actual condition of the property, and only after all that does the Tax Mapping Division update the records and encode the changes into the city's property tax system. The application then still has to pass through the Assistant City Assessor and the City Assessor for approval before it can finally be released. It is a thorough process, which makes sense given that property records need to be accurate, but it also means a single transaction can take a while and usually requires more than one visit to the office. What makes it harder for ordinary taxpayers is that all of this information, the requirements, the steps, the expected processing time, is written out in a Citizens Charter that taxpayers are simply expected to read and figure out on their own. There is no one to walk them through it unless they physically go to the office and ask, and there is no way to check the status of an ongoing application except by following up in person or over the phone. If a taxpayer is not sure whether they need a certified true copy of their tax declaration or a certification of no improvement, for instance, they usually only find out they picked the wrong document after they are already at the window.

These observations became the driving reason behind the development of this project. The researchers saw an opportunity to design a system that does not only digitize

the recording of real property tax, business tax, and market stall transactions into a single platform, but also addresses the two gaps mentioned above: the lack of a self-service inquiry channel for taxpayers, and the lack of a centralized reporting tool for treasury and assessment staff. This is where the Gemini AI-powered chatbot and the Metabase dashboard come in. The chatbot is intended to handle common taxpayer inquiries about computation, requirements, and deadlines, and can even help guide a taxpayer through figuring out which certificate or document they actually need before they step into the office, so that walk-in visits can be reserved for actual transactions instead of clarification questions. The dashboard, meanwhile, is meant to give city officials a way to monitor collection and service trends and generate reports without depending on manual spreadsheets.

1.2 Problem Statement

The Caloocan City Treasurer's Office continues to rely on manual and counter based processes for revenue collection and taxpayer transactions, resulting in long queues, delayed processing of payments and inconsistent handling of taxpayer inquiries regarding real property tax, business permits and other local fees. However, most questions and concerns are answered through face to face transactions, phone calls or social media messages. Taxpayers are often required to personally visit the city offices, secure queue numbers, submit physical documents, wait for verification and assessment, and return to claim certificates.

This process may result in long waiting lines for taxpayers especially during peak periods and increase workload for the personnel of Caloocan Treasurer's Office.

The Treasurer office also has limited tools for monitoring and analyzing the collected revenue data. Collection reports and financial records are often gathered from different sources, making the reporting process slow and time consuming for the personnel. As a result, treasury personnel may not have immediate access to updated information about revenue collection and financial performance. Also, the lack of visualization dashboards make it harder to monitor the unpaid taxes.

As technology continues to improve, the Local Government Units are expected to provide efficient, faster and accessible public services. There is a growing need for a system that can automate the treasury process. An integrated system can reduce manual work, improve the accuracy of records and make the transactions more convenient for both taxpayer and treasury personnel.

To address these problems, this study proposes the Implementation of an Intelligent Revenue Collection and Treasury Services System with Gemini AI-powered taxpayer inquiry assistance and Metabase business Intelligence dashboard for Local Government Units of Caloocan to provide a 27/7 taxpayer assistance through Artificial Intelligence for their inquiry and Business Intelligence dashboard for real time report.

Specifically, it seeks to answer the following questions:

General Problem

How can an intelligent revenue collection and treasury services system, integrated with Gemini AI-powered taxpayer inquiry assistance and a Metabase business intelligence dashboard, be developed and implemented to improve treasury operations and revenue collection efficiency for the Local Government Unit of Caloocan City?

Specific Problems

1. How can the challenges and inefficiencies in the current revenue collection and treasury processes of the Caloocan City Treasurer's Office be identified?
2. How can a Gemini AI-powered taxpayer inquiry assistant help answer common questions about real property tax, business permits, community tax, and other local fees in Caloocan City?
3. How can a Metabase business intelligence dashboard help the Caloocan City Treasurer's Office monitor revenue collection, track collection performance, and identify unpaid taxes in real time?
4. How does the developed system improve revenue collection, taxpayer service, and staff productivity compared to the current manual or semi-digital process?
5. How do the personnel of the Caloocan City Treasurer's Office and taxpayers evaluate the system in terms of:

Usability

Reliability

Effectiveness

1.3 Project Vision and Scope

To improve the revenue collection and treasury services of the City of Caloocan by developing an intelligent system that automates key treasury processes, provides taxpayers inquiry with quick AI-powered assistance and gives local government officials real-time reports for more efficient, transparent and easier for citizens to use.

1.3.1 In-Scope

Scope of the Study

This study focuses on the development and implementation of an Intelligent Revenue Collection and Treasury Services System for Caloocan City Hall. The proposed system is a web-based platform that can be accessed using computer and mobile devices through a responsive web browser to improve the management of revenue collection and treasury services by providing a centralized web-based platform. It includes modules for Real Property Tax Collection, Business Tax and Regulatory Fee Payment, Market Stall Rental and Billing, Digital Payment Integration, and Treasury Dashboard and Reports.

The study also includes the integration of Gemini AI-Powered Taxpayer Inquiry Assistance, which allows taxpayers to ask questions about tax payments, payment procedures, required documents, due dates, penalties, and other common concerns. This feature aims to provide quick and accurate responses, making it easier for taxpayers to access information without waiting for office personnel.

In addition, the system includes a Metabase Business Intelligence Dashboard that provides visual reports, charts, and revenue summaries. This feature helps treasury personnel monitor collections, analyze revenue data, identify trends, and support better decision-making. The system is intended to improve the efficiency, accuracy, transparency, and convenience of revenue collection and treasury services for both taxpayers and authorized personnel of Caloocan City Hall.

1.3.2 Out-of-Scope

This study is limited to the Revenue Collection and Treasury Services Office of Caloocan City Hall. It does not include other government functions such as budgeting, accounting, payroll management, procurement, human resource management, or other departments outside the revenue and treasury office.

The system is designed only to assist taxpayers by answering common questions through Gemini AI. It does not provide legal advice, make tax decisions, approve applications, or replace the responsibilities of treasury employees. The Metabase dashboard is used only for reporting and data visualization and does not perform financial forecasting or automatic decision-making. Any services that require approval from authorized government officials will continue to follow the existing policies and procedures of Caloocan City Hall.

1.4 Objectives and Goals

The main objective of this study is to develop and implement an Intelligent Revenue Collection and Treasury Services System that centralizes the management of revenue collection and treasury operations for the Caloocan City Treasurer's Office. Specifically, the study aims to make revenue collection processes faster, more efficient, and well-organized. The system includes a Gemini AI-Powered Taxpayer Inquiry Assistance feature that provides timely and accurate responses to common taxpayer inquiries, making information more accessible while reducing the workload of treasury personnel.

The study also includes a Metabase Business Intelligence Dashboard that generates real-time reports, charts, and data visualizations to support revenue monitoring and decision-

making. In addition, the proposed system maintains complete, accurate, and centralized revenue records to strengthen data accuracy and record management. It also provides reliable reports that promote transparency, support planning, and assist government officials in making informed decisions.

Overall, the proposed system provides faster, more convenient, and more accessible revenue collection and treasury services for both taxpayers and treasury personnel, contributing to greater efficiency, transparency, and better public service delivery.

1.4.1 General Objective

The general objective of this study is to develop and implement an Intelligent Revenue Collection and Treasury Services System with Gemini AI-Powered Taxpayer Inquiry Assistance and a Metabase Business Intelligence Dashboard for Caloocan City Hall to improve the efficiency, accuracy, transparency, and accessibility of revenue collection and treasury operations. The system also aims to provide better taxpayer support, simplify payment processes, and help treasury personnel monitor and manage revenue through organized reports and data visualization.

1.4.2 Specific Objectives

Specifically, this study aims to:

1. Identifying the revenue collection and treasury services of Caloocan City Government, including real property tax collection, business tax and regulatory fee payments, market stall rental payment, digital payment processing and financial reporting needs to serve as the basis for designing the proposed system.
2. Design and Development the Revenue Collection & Treasury Services Module consisting of Real property tax, Business tax, Market stall Rental, Billing, Digital payment,

Dashboard and Reports that can be integrated revenue collection processes and improve treasury operations

3. Integrate a Gemini AI- based Taxpayer Inquiry Assistance feature that provides automated response to taxpayer inquiries regarding taxes, fees, billing information, and payment procedures to enhance taxpayer support and reduce manual assistance workload.

4. Develop a Metabase Business Intelligence Dashboard that generates real-time analytics, revenue summary and financial reports to support data driven decision making and improve financial transparency .

5. Evaluate the developed Intelligent Revenue Collection and Treasury Services System in terms of functionality, usability, reliability, and performance efficiency based on the assessment of Caloocan City Government personnel, treasury staff and IT experts.

1.5 Significance and Relevance

This study is expected to benefit the Caloocan City Government by providing a revenue collection and treasury services system. That combines Metabase businesses intelligence dashboard with Gemini-AI powered Assistance. The proposed system aims to improve the collection of taxes, fees, and rentals while enhancing taxpayer services through artificial intelligence and data analytics. The following are the beneficiaries of the study:

For Caloocan City Government: The system will enhance revenue collection, improve financial transparency, strengthen treasury operations, and support better public service delivery through digital transformation.

For Business Owners:

The system will improve the payment of business taxes and regulatory fees by providing a centralized platform for billing, payment tracking, and transaction records, resulting in a more efficient payment process.

For Market Stall Tenants: The system will enable easier monitoring of rental fees, billing schedules, and payment records, helping tenants stay informed about their financial obligations.

For Treasury Personnel and Revenue Collectors:

The system will automate revenue collection processes, reduce manual paperwork, improve record management, and facilitate the generation of accurate financial reports, thereby increasing operational efficiency.

For the taxpayers:

The system will provide taxpayers with a more convenient and accessible way to inquire about taxes, fees, billing information, and payment procedures through the Gemini AI-powered inquiry assistant. It will also offer digital payment options, reducing the time and effort required to complete transactions.

For Decision-Makers and Administrators: The Metabase Business Intelligence Dashboard will provide real-time reports and analytics that can assist in monitoring revenue performance, identifying trends, and makin.

For Future Researchers: This study may serve as a reference for future research related to intelligent government systems, artificial intelligence applications, digital payment integration, and business intelligence solutions in public administration.

1.5.1 Theoretical Significance

This study is anchored on the following theoretical frameworks:

According to Solomon et al. (2024), organizations can improve operational performance and decision-making by integrating advanced analytics, interactive intelligence dashboards, and artificial intelligence (AI)-powered predictive models into their information systems. This framework emphasizes the value of data-driven technologies in enhancing organizational efficiency and supporting strategic planning.

The core concept of this framework is that advanced analytics transforms raw operational data into meaningful insights, while interactive dashboards present information in a clear and accessible manner. AI-powered predictive models further strengthen decision-making by identifying trends, forecasting outcomes, and enabling organizations to respond proactively to operational challenges.

This theory supports the proposed Revenue Collection and Treasury Services System because it incorporates a Metabase Business Intelligence dashboard to visualize financial and collection data through interactive reports and analytics. Additionally, the integration of Google Gemini AI enables intelligent data interpretation and automated assistance, helping treasury personnel analyze revenue trends, monitor collections, and make informed decisions for improved financial management in the Caloocan City Government.

Data Intelligence Implementation Theory: Merhi (2021) proposed that successful implementation of data intelligence **in the public sector depends on** critical **factors** such as data quality, organizational readiness, technological capability, and management support. The study highlights that effective data intelligence enables government agencies to improve transparency, efficiency, and evidence-based decision-making.

The central concept of this theory is that public organizations achieve better performance when reliable data is collected, processed, and transformed into actionable information through business intelligence tools. Data intelligence strengthens operational monitoring and allows government officials to make timely and accurate decisions based on real-time information.

This theory is relevant to the proposed system because the Metabase BI dashboard provides treasury personnel with real-time visual reports on tax collections, business permits, market stall payments, and other financial transactions. By transforming collected data into meaningful insights, the system supports efficient revenue monitoring, improves transparency, and enhances decision-making within the Caloocan City Government.

Sustainable AI Chatbot Adoption Theory: Rathnayake et al. (2025) explained that citizens are more likely to adopt AI chatbots in government services when the systems demonstrate trustworthiness, transparency, reliability, and policy compliance. Their study emphasizes that user trust is a key factor in encouraging the successful implementation of AI-powered public services.

The theory suggests that AI chatbots become more effective when they provide accurate information, respond consistently to user inquiries, and maintain transparency in their interactions. Establishing trust encourages citizens to engage with digital government services while improving accessibility and service efficiency.

This theory supports the proposed system because it integrates a Google Gemini AI chatbot that assists taxpayers by answering inquiries related to tax payments, business

permits, market stall rentals, payment procedures, and treasury services. By providing accurate and reliable responses, the chatbot enhances user experience, promotes trust in digital government services, and encourages citizens to utilize the online revenue collection system of the Caloocan City Government.

Citizens' Acceptance of Artificial Intelligence in Public Services: Horvath et al. 2023) examined the factors influencing citizens' acceptance of artificial intelligence in public services and found that public trust, perceived fairness, transparency, and service efficiency significantly affect the willingness of citizens to use AI-enabled government systems.

The main concept of this study is that citizens are more likely to accept AI technologies when they believe these systems improve service delivery while maintaining fairness, accountability, and transparency. AI should complement government services by increasing efficiency without compromising public confidence.

This theory is applicable to the proposed Revenue Collection and Treasury Services System because the integration of Google Gemini AI aims to improve the quality of government services by providing automated assistance, answering taxpayer inquiries, and guiding users throughout the revenue collection process. Together with the Metabase BI dashboard, the system enhances transparency, operational efficiency, and informed decision-making, ultimately improving the delivery of treasury services in the Caloocan City Government.

1.5.2 Practical Significance

The following stakeholders in Caloocan City's local government and community stand to gain

immediately from the deployment of this system:

To the Caloocan City Treasurer's Office and Staff:

The system automates routine tasks such as billing and assessment, greatly reducing the amount of work that has to be done by hand. Staff do not have to repeat the same questions repeatedly. It allows them to concentrate on difficult jobs, reduce processing backlogs and prevent long lines.

To LGU Executives and Local Policymakers:

City administrators, the City Treasurer and local executives are getting clear, real-time visualizations of data. They don't have to manually compile reports anymore, which can take weeks. Instead they are able to instantly view collection trends, daily performance and identify areas with significant unpaid taxes. This enables them to make faster and more calculated financial decisions.

To the Local Government Unit (LGU) as a Whole:

Improving the speed of gathering directly raises local income. With higher collection rates, Caloocan City can more easily access the money it needs to fund public services, construction upgrades and community projects. This leads to greater economic freedom locally.

To the Taxpayers (Citizens and Business Owners of Caloocan City):

People who pay taxes have a much easier time. They don't have to wait in line or on

social media for hours to check a fee anymore. They can get live, exact help from the AI helper 24/7 when they need it with things like real property taxes or business permits.

1.6 Definition of Terms

Business Intelligence Dashboard A business intelligence dashboard refers to the visual reporting interface that presents revenue collection data through charts, graphs, and key performance indicators. It enables the Caloocan City Treasurer's Office to monitor collection performance, identify unpaid taxes, and support data-driven decision-making using a metabase.

Caloocan City Treasurer's Office

The Caloocan City treasurer's office refers to the local government office responsible for collecting local taxes, fees, charges, and other revenues, as well as managing the treasury functions that serve as the development and evaluation of the proposed system.

Gemini AI-Powered Taxpayer Inquiry Assistance

Gemini AI-Powered taxpayer inquiry assistance refers to the artificial intelligence feature integrated into the proposed system that provides automated responses to taxpayer inquiries regarding real property tax, business permits, community tax, local fees, payment procedures, and other treasury-related concerns.

Intelligent Revenue Collection and Treasury Services System

The Intelligent Revenue Collection and Treasury Services System refers to the proposed web-based information system designed to automate and improve revenue collection and treasury operations through digital transaction processing, artificial intelligence-assisted taxpayer support, and business intelligence reporting.

Local Government Unit (LGU)

The Local Government Unit refers to the City Government of Caloocan, particularly the offices involved in revenue collection and treasury operations that will utilize the proposed system.

Metabase

Metabase refers to the open source business intelligence and data visualization platform integrated into the proposed system to generate interactive dashboards, reports, and analytics for monitoring revenue collection and treasury performance.

Real Property Tax

Real property tax refers to the local tax imposed on land, buildings, machinery, and other real properties within the jurisdiction of the local government unit, which taxpayers may inquire about through the AI-powered assistance feature.

Revenue Collection

Revenue collection refers to the process of receiving, recording, and managing payments from taxpayers for local taxes, fees, permits, and other government charges administered by the Caloocan City Treasurer's Office.

Taxpayer

A taxpayer refers to any individual, business establishment, or juridical entity that is required to pay local taxes, fees, permits, or other government charges and who uses the proposed system for transactions or inquiries.

Taxpayer Inquiry

A Taxpayer inquiry refers to any question, request for information, or concern submitted by taxpayers regarding taxes, permits, payment requirements, due dates, penalties, and other treasury-related services through the proposed system.

Treasury Services

Treasury Services refers to the functions performed by the Caloocan City Treasurer's Office related to the assessment, collection, recording, monitoring, and reporting of local government revenues.

Unpaid Taxes

Unpaid taxes refer to tax obligation that remain unsettled after their prescribed due dates. These records are monitored through the metabase business intelligence dashboard to support collection monitoring and follow-up activities.

Web-Based System

a web-based system refers to the online application that allows authorized personnel and taxpayers to access the proposed revenue collection and treasury services system using an internet-connected web browser.

1.7 Structure of the Document

Chapter	1:	Introduction
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Provides an overview of the project, including the background and motivation, problem statement, project vision and scope, general and specific objectives, as well as the theoretical and practical significance of the study to the City of Caloocan.

Chapter	2: Related Literature, Studies, and Technical background
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reviewing current research, regional laws, and international systems that are pertinent to

the project. Republic Act No. 7160, the usage of Gemini AI for automated customer help, the construction of e-governance frameworks, and the practical use of Metabase Business Intelligence dashboards are all part of it.

Chapter 3: Methodology and Project Management

Describes the technical approach and software engineering process followed to build the system. This describes the architecture of the system, the schema of the database design, the integration of the Gemini AI-powered assistant, the setup of the metabase dashboards and the aligned evaluation instrument.

Chapter 4: System Design, Development, And Implementation

Presents the empirical results of the project actual implementation. It measures the system's deployment performance, the responsiveness of the AI inquiry assistant, the visualization capabilities of the Metabase dashboards, and the data collected via user acceptance testing (UAT).

Chapter 5: Summary, Conclusion, And Recommendations

The study concludes with a concluding assessment of the system's impact on the processing of LGU revenue and a summary of the development findings. Future functional enhancements and additional research developments are suggested.